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|  <b>Reliable</b><br>INSTITUTE<br>Unleashing Potential | <b>MATHEMATICS</b><br><b>REVISION PLAN-3</b><br><b>Target : JEE (Advanced)-2023</b> | <b>DPP</b><br>DAILY PRACTICE PROBLEM<br><b>DPP NO. # 05 (Advanced)</b> |
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| Max. Time : 129 Minutes                                                                        | Total Marks-166 |
|------------------------------------------------------------------------------------------------|-----------------|
| Q. 01 to 06    Comprehension    ('-1' negative marking)    [3 Marks,3 Min.][18,18]             |                 |
| Q. 07 to 26    Multiple choice objective    ('-1' negative marking)    [4 Marks,3 Min.][80,60] |                 |
| Q. 27 to 28    Match the column    (no negative marking)    [4 Marks,3 min.][8,6]              |                 |
| Q. 29 to 43    Numerical Type    ('-1' negative marking)    [4 Marks,3 Min.][60,45]            |                 |

**DPP Syllabus :** Straight Lines, Circles and Conics Section

### Comprehension # 1 (Q.No. 1 to 2)

Let A,B,C be three sets such that

$$A = \{ (x,y) : x = 1 + r \cos \theta, y = r \sin \theta, r = 2, \theta \in \mathbb{R} \},$$

$$B = \{ (x,y) : x = -1 + r \cos \theta, y = r \sin \theta, \theta = \pi/4, r \in \mathbb{R} \},$$

$$C = \{ (x,y) : (x - x_2)^2 + (y - y_2)^2 \leq a^2, a \in \mathbb{R} \}$$

Let  $C_1$  and  $C_2$  are curves represented by sets A and B while R is the region represented by set C. Also  $P(x_1, y_1)$  and  $Q(x_2, y_2)$  are points such that  $P, Q \in A \cap B$  and  $x_1 < 0, x_2 > 0$

- The point of intersection of tangents drawn to the curve  $C_1$  at P and Q is  
(A) (-1, 0)                      (B) (2, 3)                      (C) (-1, 2)                      (D) (1, -2)
- If minimum value of 'a' for which  $A \cap C = A$  is ' $\alpha$ ' then length of curve  $C_2$  lying in R corresponding to  $a = \alpha$  is  
(A) 2                                  (B)  $\sqrt{58}$                       (C) 8                                  (D)  $\sqrt{57}$

### Comprehension # 2 (Q.No. 3 to 4)

If the tangents to parabola at P & Q meet in T, then triangle  $\Delta SPT$  is similar to triangle  $\Delta STQ$  (where 'S' being the focus of parabola). Hence the circle through point of intersection of three tangents on the parabola passes through its focus.

- The co-ordinates of focus of a parabola which touches  $x = 0, y = 0, x + y = 1$  and  $y = x - 2$   
(A)  $\left(\frac{6}{5}, -\frac{2}{5}\right)$                       (B)  $\left(-\frac{6}{5}, -\frac{2}{5}\right)$                       (C)  $\left(\frac{3}{5}, \frac{1}{5}\right)$                       (D)  $\left(\frac{6}{5}, \frac{2}{5}\right)$
- If three parabolas touch all the lines  $x = 0, y = 0$  and  $x + y = 2$ , then maximum area of the triangle formed by joining their foci.  
(A)  $\frac{\sqrt{3}}{2}$                                   (B)  $\frac{3\sqrt{3}}{2}$                                   (C)  $3\sqrt{2}$                                   (D)  $\frac{27}{4}$

### Comprehension # 3 (Q.No. 5 to 6)

A point P moves such that the sum of the slope of the normals drawn from it to the hyperbola  $xy = 16$  is equal to the sum of ordinate of feet of normals. If the locus of 'P' is curve 'C' then

- Equation of curve C is  
(A)  $x^2 = 2y$                       (B)  $x^2 = 16y$                       (C)  $y^2 = 2x$                       (D)  $y^2 = 16x$

6. Let image of curve  $C$  about the line  $x - y = 0$  is  $C'$ . Then how many number of triangle can be inscribed in the rectangular hyperbola  $xy = c^2$ , whose sides all touch the curve  $C'$ ?  
 (A) 4 (B) 3 (C) 2 (D) infinite
7. Consider a  $\Delta ABC$  whose sides  $AB$ ,  $BC$  and  $CA$  are represented by the straight lines  $2x + y = 0$ ,  $x + py = q$  and  $x - y = 3$  respectively. The point  $P$  is  $(2, 3)$ .  
 (A) If  $P$  is the centroid, then value of  $(p + q) = 74$   
 (B) If  $P$  is the orthocentre, then value of  $(p + q) = 50$   
 (C) If  $P$  is the circumcentre, then value of  $(p + q) = 47$   
 (D) If  $P$  is the orthocentre, then value of  $(p + q) = 47$
8. The orthocentre and centroid of a  $\Delta ABC$  are  $(2, 3)$  and  $\left(\frac{4}{3}, \frac{1}{3}\right)$  respectively. If equation of side  $BC$  is  $x - y = 0$ , then  
 (A) Orthocentre of  $\Delta BCH$  is  $(4, 1)$   
 (B) Orthocentre of  $\Delta BCH$  is  $(-1, 2)$   
 (C) Equation of circum-circle of  $\Delta ABC$  is  $x^2 + y^2 - 2x + 2y - 11 = 0$   
 (D) Equation of circum-circle of  $\Delta ABC$  is  $x^2 + y^2 - 2x + 2y = 0$
9. If one of the lines in  $2x^2 + axy + 3y^2 = 0$  coincide with one of those given by  $4x^2 + bxy - 6y^2 = 0$  and the other lines represented by them be perpendicular then which of the following may be true ?  
 (A)  $a + b = 7$  (B)  $a + b = -6$  (C)  $a - b = -3$  (D)  $a - b = 2$
10. A circle is made to pass through the point  $(1, 2)$  and touching the lines  $7x - y = 5$  and  $x + y + 13 = 0$ . The area of the quadrilateral formed by these tangents and corresponding normal drawn to the circle can be  
 (A) 100 (B) 200 (C) 300 (D) 400
11. The circle  $x^2 + y^2 + 2gx + 2fy + c = 0$  intersects the  $x$ -axis as well as the  $y$ -axis at two distinct points. If the lines  $x - 2y = 5$  &  $2x - y = 4$  cut the circle orthogonally, then 'c' can be  
 (A) 0 (B)  $\frac{1}{2}$  (C) 1 (D) 2
12. Two circles of radii  $R$  and  $r$ , where  $R > r$  touch each other externally. The radius of circle which touches both of them externally and also their direct common tangent is  
 (A)  $R$  (B)  $\frac{R}{R+r}$  (C)  $\frac{Rr}{(\sqrt{R} + \sqrt{r})^2}$  (D)  $\frac{Rr}{(\sqrt{R} - \sqrt{r})^2}$
13. A variable circle passes through origin  $O$  and cuts off intercepts  $OA$  and  $OB$  on  $x$ -axis and  $y$ -axis respectively such that  $OA + OB = 1$ . If circle always passes through a fixed point other than origin  $O$ . Then coordinates of the fixed point can be  
 (A)  $\left(\frac{1}{2}, \frac{1}{2}\right)$  (B)  $\left(-\frac{1}{2}, \frac{1}{2}\right)$  (C)  $\left(\frac{1}{2}, -\frac{1}{2}\right)$  (D)  $\left(-\frac{1}{2}, -\frac{1}{2}\right)$
14. The conic represented by the following equation  $\sqrt{px} + \sqrt{qy} = 1$ , where  $p, q > 0$   
 (A) Parabola for  $p = q = 2$  (B) Ellipse for  $p = q = \frac{1}{2}$   
 (C) Hyperbola for  $p = q = 3$  (D) Unique conic for all  $p, q > 0$

15. From a point P, tangents PQ and PR are drawn to the parabola  $y^2 = 4ax$ . Then centroid of  $\Delta PQR$   
 (A) lies inside parabola (B) lies outside parabola  
 (C) has ordinate same as P (D) has abscissa same as P
16. Consider two parabola,  $P_1$  &  $P_2$  with vertices  $V_1(0, 4)$  &  $V_2(6, 0)$  respectively.  $P_1$  &  $P_2$  are tangent to each other at point P and have their axis of symmetry parallel to y-axis. Then  
 (A) Either sum of latus rectum or difference of latus rectum equal to 9  
 (B) If  $P_1$  &  $P_2$  are equal, then length of their latus rectum equal to  $9/2$   
 (C) If  $P_1$  &  $P_2$  are equal, then length of their latus rectum equal to 9  
 (D) P is (3, 2), if  $P_1$  &  $P_2$  are equal parabola
17. If the normal chords at points P & Q on the parabola  $y^2 = 4ax$  meets again on the parabola at R, then the locus of circumcentre of  $\Delta PQR$  is  
 (A) A parabola with focus  $\left(\frac{9a}{8}, 0\right)$  (B) A parabola with vertex (0, a)  
 (C) A parabola with directrix  $x = \frac{7a}{8}$  (D) Parabola passing through (3a, -a)
18. Focus and vertex of a parabola are A(5, 2) and B(8, 6) respectively. If P and Q are two points on the parabola such that the tangents meet at T(11, 10), then  
 (A) P and Q are mirror images of each other in the line  $4x - 3y = 14$   
 (B) Area of quadrilateral formed by tangent and normal at P and Q is 400 sq. units  
 (C) Area of quadrilateral formed by tangent and normal at P and Q is 200 sq. units  
 (D) P and Q are the extrimities of latus rectum of this parabola
19. Which of the following is/are true?  
 (A) There are infinite positive integral values of 'a' for which in  $(13x - 1)^2 + (13y - 2)^2 = \left(\frac{5x + 12y - 1}{a}\right)^2$  represents an ellipse  
 (B) The minimum distance of a point (1, 2) from the ellipse  $4x^2 + 9y^2 + 8x - 36y + 4 = 0$  is 1  
 (C) If from a point P(0,  $\alpha$ ) two normal other than the axes are drawn to the ellipse  $\frac{x^2}{25} + \frac{y^2}{16} = 1$  then  $|\alpha| < 9/4$   
 (D) If the length of the latus rectum of an ellipse is  $1/3$  of its major axis, then its eccentricity is  $\frac{1}{\sqrt{3}}$
20. Let tangent at point 'P' on ellipse cuts directrix corresponding to focus  $S_1$  and  $S_2$  of ellipse at  $D_1$  and  $D_2$  respectively. This tangent cuts auxiliary circle at  $F_1$  and  $F_2$  (where  $F_1D_1 < F_2D_1$ ). Lines  $D_1S_1$  and  $D_2S_2$  meets at Q. If  $F_1F_2$  = length of semi major axis, then  
 (A)  $D_1Q = D_2Q$  (B)  $D_1Q \neq D_2Q$  (C)  $\angle PD_1S_1 = 30^\circ$  (D)  $\angle PD_2S_2 = 60^\circ$

21. The line  $3x - 4y - 1 = 0$  touches the ellipse at point  $P \equiv (-1, -1)$ , whose one focus is at origin and length of semi-major axis is  $13\sqrt{2}$ . Then which of the following is/are correct?
- (A) Point  $\left(-24, -\frac{27}{2}\right)$  lies outside the ellipse
- (B) Eccentricity of the ellipse is  $\sqrt{\frac{337}{338}}$
- (C) If normal at P intersect major and minor axes at G & g, then  $PG.Pg = 50$
- (D) Equation of minor axis is  $16x + 9y + 337 = 0$
22. Major and minor axis of an ellipse are 8 and 6 respectively. Initially it touches positive x & y axis and line joining the two foci is parallel to x-axis. It then rotates in anti-clockwise sense, always touching both the positive coordinate axes, and the rotation stops when the line joining their foci is vertical for the first time. 'C' is centre of ellipse & 'O' is origin, then
- (A) Locus of 'C' is the complete portion of  $x^2 + y^2 = 25$  lying in the first quadrant
- (B) Locus of 'C' is part of the circle  $x^2 + y^2 = 25$
- (C) Total distance covered by 'C' is  $5 \tan^{-1}\left(\frac{7}{24}\right)$
- (D) Initial & final positions of 'C' lies on the curve  $xy = 12$
23. From any point on the line  $(t + 1)(x + y) = 1$  (where t is a parameter,  $t \neq -1$ ) tangents are drawn to the ellipse  $x^2 + 4y^2 = \frac{1}{4}$ . In this case the chord of contact will pass through a fixed point for a particular 't'. Then which of following number(s) contained in the range of 't' so that the point of concurrency will lie inside the ellipse
- (A) 0 (B) -1 (C) 2 (D)  $\frac{1}{2}$
24. A variable chords of the hyperbola  $\frac{x^2}{9} - \frac{y^2}{\lambda^2} = 1$  subtends a right angle at centre of the hyperbola. If this chord touches A fixed circle concentric with the hyperbola, then
- (A) radius of the circle is an irrational number for  $\lambda = 2$
- (B) the circle passes through  $\left(\frac{1}{2}, \frac{\sqrt{7}}{2}\right)$  for  $\lambda = 3$
- (C) the circle passes through  $\left(\sqrt{3}, \sqrt{\frac{123}{7}}\right)$  for  $\lambda = 4$
- (D) radius of the circle is a rational number for  $\lambda = 5$
25. If P is a point on a hyperbola, then
- (A) locus of excentre of the circle described opposite to  $\angle P$  for  $\Delta PSS'$  (S, S' are foci) is tangent at vertex
- (B) locus of excentre of the circle described opposite to  $\angle S'$  is hyperbola
- (C) locus of excentre of the circle described opposite to  $\angle P$  for  $\Delta PSS'$  (S, S' are foci) is hyperbola
- (D) locus of excentre of the circle described opposite to  $\angle S$  is tangent at vertex

26. Let  $a$  and  $b$  be positive real numbers such that  $a > 1$  and  $b < a$ . Let  $P$  be a point in the first quadrant which lies on the hyperbola  $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ . Suppose, the tangent to the hyperbola at  $P$  passes through the point  $(1, 0)$  and, the normal to the hyperbola at  $P$  cut-off equal intercepts on the coordinate axes. Let  $\Delta$  denote the area of the triangle formed by the tangent at  $P$ , the normal at  $P$  and the  $X$ -axis, if  $e$  denotes the eccentricity of the hyperbola, then which of the following statements is/are true?

- (A)  $1 < e < \sqrt{2}$       (B)  $\sqrt{2} < e < 2$       (C)  $\Delta = a^4$       (D)  $\Delta = b^4$

27.  $L_1 : 3x + y = 0$

$$L_2 : 4x + 3y + 5 = 0$$

Consider family of straight lines.

$$L_1 + \lambda L_2 = 0 ; \lambda \in \text{parameter.}$$

Match List-I with List-II .

**List-I**

- (A) Thrice of absolute value of  $y$ -intercept of line having

$$\text{slope } -\frac{[\pi+1]}{[e+1]}$$

(where  $[.]$  represent greatest integer function)

- (B) Square of maximum distance of point  $(2, -6)$  from any member of family is

- (C) If locus of foot of perpendicular drawn from  $(0,0)$  to any member of family is  $x^2 + y^2 + ax + by = 0$ , then  $(a + b)$  is

- (D) Let  $L_3$  is a member of family of lines passes through point of intersection of  $L_1$  and  $L_2$  which passes through  $P\left(-\frac{5}{4}, 0\right)$ , then distance of  $L_3$  from  $Q(4,3)$  is

**List-II**

- (p) 2

- (q) 6

- (r) 5

- (s) 10

28. Let  $S_1$  and  $S_2$  are two unit radius circles with centre at  $C_1(0,0)$  and  $C_2(1,0)$  respectively. Let  $S_3$  is another circle of unit radius, passes through  $C_1$  and  $C_2$  and its centre is above the  $x$ -axis. The equation of common tangent to  $S_1$  and  $S_3$  which does not pass through  $S_2$  is  $ax + by + 2 = 0$ .

Match List-I with List-II.

**List-I**

- (A) The ordinate of centre of circle  $S_3$  is

- (B) The value of  $a$  is equal to

- (C) The value of  $(a^2 - b)$  is equal to

- (D) The abscissa of centre of circle  $S_3$  is

**List-II**

- (p)  $\frac{1}{2}$

- (q)  $\frac{\sqrt{3}}{2}$

- (r)  $\sqrt{3}$

- (s) 4

29. A  $(-a, 0)$  & B  $(a, 0)$  are fixed points and C is a point which divides AB in a constant ratio,  $\tan \alpha: 1$ . If the locus of a point P on which the line segments AC and CB subtend equal angles is  $x^2 + y^2 + kax \sec(m\alpha) + a^2 = 0$ ,  $2k - m =$
30. Triangle ABC having side lengths  $a = 8$ ,  $b = 6$  and  $c = 10$  slides on the coordinate axes with vertex A on x-axis and B on y-axis. Locus of vertex C is a straight line of slope equal to
31. The square of distance between the feet of perpendiculars drawn from a point P  $(1, 0)$  on to the pair,  $x^2 + 8xy + y^2 = 0$  is,
32. Consider a line pair  $ax^2 + 3xy - 2y^2 - 5x + 5y + c = 0$  representing perpendicular lines intersecting each other at C and forming a  $\Delta ABC$  with the x-axis. If the circle  $x^2 + y^2 - 4y + k = 0$  is orthogonal with the circumcircle of the  $\Delta ABC$ , then k equals
33.  $C_1$  is a fixed circle and  $C_2$  is the director circle of  $C_1$ . A third circle  $C_3$  passes through the centre of  $C_1$ . It cuts circle  $C_1$  at points A, B and circle  $C_2$  at P, Q. If PQ touches circle  $C_1$  then find the value of  $\left(\frac{AB}{PQ}\right)^2$
34. Consider a series of n concentric circles  $C_1, C_2, C_3, \dots, C_n$  with radii  $r_1, r_2, r_3, \dots, r_n$  respectively satisfying  $r_1 > r_2 > r_3 > \dots > r_n$  with  $r_1 = 10$ . The circles are such that the chord of contact of tangents from any point on  $C_i$  to  $C_{i+1}$  is a tangent to  $C_{i+2}$  ( $i = 1, 2, 3, \dots$ ). Find the value of  $\lim_{n \rightarrow \infty} \sum_{i=1}^n r_i$  if angle between tangents from any point of  $C_1$  to  $C_2$  is  $\frac{\pi}{3}$
35. Given a point P(a, b) in the first quadrant. Circles are drawn through P touching the coordinate axes such that the length of common chord of these circles is maximum. If  $\frac{a}{b} = p \pm q\sqrt{2}$ , then find  $p + q$
36. Two equal parabolas of latusrectum 8 units have the same focus and their axes are at right angle. A normal to the parabola is perpendicular to a normal to other parabola. Then the locus of the point of intersection of those normals is another parabola whose square of length of latusrectum equals
37. If S and S' are two foci of the ellipse  $16x^2 + 25y^2 = 400$  and PSQ is a focal chord such that  $SP = 16$ , then find value of  $9S'Q$ .
38. Rectangle ABCD has area 200. An ellipse with area  $200\pi$  passes through A & C and has foci at B & D. If the perimeter of the rectangle is P, then the value of  $P/20$  is
39. A coplanar beam of light emerging from a point source has the equation  $\lambda x - y + 2(1 + \lambda) = 0$ ,  $\lambda \in \mathbb{R}$ , the rays of the beam strike an elliptical surface and gets reflected. The reflected rays form another convergent beam having equation  $\mu x - y + 2(1 - \mu) = 0$ ,  $\mu \in \mathbb{R}$ . Further it is found that the foot of the perpendicular from the point  $(2, 2)$  upon any tangent to the ellipse lies on the circle  $x^2 + y^2 - 4y - 5 = 0$ . The locus of middle point of focal chord of this ellipse is another ellipse whose eccentricity is equal to

40. If the normals at  $(x_r, y_r)$ , where  $r = 1, 2, 3, 4$  to the ellipse  $\frac{x^2}{2023} + \frac{y^2}{2022} = 1$  are concurrent, then value of  $(x_1 + x_2 + x_3 + x_4) \left( \frac{1}{x_1} + \frac{1}{x_2} + \frac{1}{x_3} + \frac{1}{x_4} \right)$  is equal to
41. Let PQ and RS be two perpendicular focal chords of a rectangular hyperbola, which are not parallel to its axes, then  $\frac{PQ}{RS}$  is equal to
42. The tangent to the hyperbola  $xy = 8$  at the point 'P' intersects the x-axis at T and the y-axis at T'. The normal to hyperbola at P intersects the x-axis at N and y-axis at N'. The area of  $\Delta PNT$  &  $\Delta PN'T'$  are  $\Delta_1$  and  $\Delta_2$  respectively, then the value of  $\frac{1}{\Delta_1} + \frac{1}{\Delta_2}$  is
43. Curves  $(x - 1)(y - 2) = 5$  and  $(x - 1)^2 + (y + 2)^2 = r^2$  intersect at four points A, B, C, D. If centroid of  $\Delta ABC$  lies on the line  $y = 3x - 4$  then locus of D is a straight line  $y = mx$ . Find m.

## ANSWER KEY

1. (C) 2. (C) 3. (D) 4. (B) 5. (B) 6. (D) 7. (ABC)
8. (AC) 9. (AC) 10. (AD) 11. (AB) 12. (CD) 13. (ABCD)
14. (AD) 15. (AC) 16. (ABD) 17. (ACD) 18. (ACD) 19. (ABC) 20. (AC)
21. (BCD) 22. (BCD) 23. (AD) 24. (CD) 25. (AB) 26. (AD)
27.  $(A) \rightarrow (r); (B) \rightarrow (s); (C) \rightarrow (p); (D) \rightarrow (q)$
28.  $(A) \rightarrow (q); (B) \rightarrow (r); (C) \rightarrow (s); (D) \rightarrow (p)$
29. (2) 30. (0.75) 31. (0.93 or 0.94) 32. (1.50) 33. (0.75) 34. (20)
35. (5) 36. (8) 37. (74) 38. (4) 39. (0.66 or 0.67) 40. (4)
41. (1) 42. (0.25) 43. (3)